# REWARD ERRORS BECOME PERSISTENT STATE: WRITE-TIME CAUSALITY AND TRANSPORT BOUNDARIES IN AGENT MEMORY

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## ABSTRACT

Many external-memory agents use terminal feedback both to score an episode and to choose how its trajectory is written into persistent memory. We isolate this write-time channel in released ReasoningBank/WebArena Shopping trajectories by holding each source experience byte-identical and flipping only the success/failure reflection branch. The original four complete pairs all diverge, and a preregistered breadth replication adds 16/16 divergent pairs across 11 intent-template families, yielding 20 complete paired sources. On eight disjoint controls, reward-mode divergence exceeds stronger same-mode semantic-preserving rewording (0.700 vs. 0.595; exact $p = 0.0078$). A forced-memory $4 \times 4$ terminal replication shows conditional sensitivity (mean $|\Delta| = 0.15625$, $p = 0.00074$), but exact retriever replay shows that none of those four future tests retrieves its source memory under the original bank. Expanding to 20 sources yields 36 deterministic retrieval-hit futures. Their 288 success/failure rollouts give mean $|\Delta| = 0.02083$ ($p = 0.4289$); matched no-memory and raw-trajectory controls also miss the 0.15 practical floor. A preregistered 432-call first-action probe localizes this attenuation: mean success/failure action-distribution TV is 0.06944 ($p = 0.5801$), with zero modal branch differences across 36 states. Thus reward errors robustly change what is written, while branch-specific uptake and realized transport are separate gates.

## 1 INTRODUCTION

External-memory agents can turn the end of one episode into persistent state for the next. A completed trajectory is summarized, stored, retrieved, and injected into later context, often without any parameter update. When terminal feedback selects how that summary is written, the evaluator is therefore part of the state-update mechanism rather than only a scorer. An erroneous label can cause the same experience to be committed through a different reflection objective and remain available to future decisions.

Several neighboring literatures establish pieces of this pipeline without separating its stages. JudgeBench and RewardBench document failures of LLM judges and reward models (Tan et al., 2025; Lambert et al., 2024). ReasoningBank and Reflexion turn success/failure feedback into reusable textual memory (Ouyang et al., 2026; Shinn et al., 2023); Mem-$\alpha$ and AttriMem learn memory-construction policies from outcome or process reward (Wang et al., 2025; Li et al., 2026). Memory Reward Inflation and RoMeRL study reward-driven valuation and reuse of memories that already exist (Asadolahi et al., 2026; Yang et al., 2026), while recent fragility work measures run variance and task-order sensitivity (Ye et al., 2026). We therefore do not claim judge unreliability, feedback-driven memory construction, memory-reward amplification, or self-improvement variance as new. The missing identification object is earlier: under a fixed construction policy and byte-identical experience, can a terminal-label error select a different durable memory state? A second, distinct question is whether that written difference is actually exposed and behaviorally transported by the released retrieval/interface stack.

We first isolate the write boundary. For each source episode, task and trajectory are held byte-for-byte identical while the terminal label counterfactually selects the released success- versus failure-reflection branch. Because those branches necessarily use different instructions, generic prompt sensitivity is the strongest reduction: on eight disjoint trajectories we compare the reward-mode intervention against semantic-preserving rewordings within the same mode whose lexical perturbation is deliberately larger. The original intervention yields four complete pairs, all divergent. We then preregister a breadth replication over 16 additional sources, balanced 8/8 by original benchmark outcome and spanning 11 intent-template families. A 2,200-token parent run is execution-censored in three failure-branch writes; one uniform 4,096-token repair regenerates all 32 units fresh and completes 16/16 pairs. Combined with the original sample, all 20 complete paired sources change both memory content and title sets. Thus the upstream write-time effect is no longer a four-source phenomenon.

We next separate *intervention sensitivity* from *realized transport*. Under a forced memory swap, the original four complete source pairs alter 7/12 aligned next actions. A same-support $4 \times 4$ terminal replication after an initial non-pass keeps the same support and dual gate, increases only replication depth, and yields mean absolute success-rate difference 0.15625 with permutation $p = 0.00074$. This establishes that the paired persistent states *can* alter later outcomes when experimentally injected under matched evidence. However, exact replay of the released ReasoningBank retriever changes the interpretation of that result: with the original four-memory bank, only 8/188 held-out Shopping tasks pass the released all-MiniLM-L6-v2 top-1, 0.3 threshold, and none of the four forced-swap future tasks would retrieve those memories.

We therefore expand the source bank before testing ecological transport. With 20 source descriptions, exact released-retriever exposure rises to 125/172 held-out Shopping tasks (72.7%), spanning 39 intent-template families. Before any terminal outcomes, we freeze the 36 tasks that additionally have released trajectories and deterministic offline evaluators. For each task, the retriever itself selects the source memory; we then swap only that source's success- versus failure-conditioned memory and serialize it with the released ReasoningBank human-memory wrapper. All 288 terminal calls complete. The broad retrieval-matched contrast is small: mean $|\Delta| = 0.02083$ with $p = 0.4289$, and 34/36 tasks have zero branch contrast. Eighteen of those are joint 0/0 floors and sixteen joint 1/1 ceilings. A matched 144-call no-memory arm gives mean memory-presence effect 0.04514; its omnibus $p = 0.00147$ is statistically detectable but misses the unchanged 0.15 practical-effect floor.

Three follow-ups localize the attenuation. A matched 144-call raw writer-input trajectory arm gives mean rewrite-versus-raw effect 0.04514 ($p = 0.00775$), again below the 0.15 floor. A zero-call post-hoc fractional-reference re-score reduces joint S/F floors from 18 to 10 but leaves mean S/F separation at 0.01951 overall; endpoint discretization therefore hides some cell-level variation but not a large branch effect. Finally, a preregistered 432-call first-action probe reuses the same 36 native-hit tasks, retriever-selected sources, and memory wrappers but freezes the released step-1 browser state before any terminal outcome. Mean success/failure first-action TV is only 0.06944 ($p = 0.5801$) against the inherited 0.20 process floor, and the modal first action differs in 0/36 states. A descriptive memory-versus-no-memory TV of 0.17014 with six modal shifts suggests generic memory presence can affect procedure, while reward-branch-specific uptake is weak.

These results expose a useful boundary rather than a contradiction. Reward-conditioned writing robustly changes persistent state; a forced swap proves that some such state differences can matter; but native branch-specific transport attenuates before or at policy uptake on most held-out tasks. The data therefore separate *write divergence*, *retrieval exposure*, *branch-specific uptake*, and *terminal transport*. Endpoint headroom and representation remain secondary gates rather than complete explanations. A GLM-5.3 writer independently reinforces the upstream/downstream split: all four paired memories diverge, while its terminal replication misses the practical-effect floor. We also preregister a DeepSeek-V4-Flash downstream-policy transfer on the same 36 tasks, but both the 900-token parent and the single allowed 2,200-token output-cap repair terminate on the first frozen unit with provider length censoring and no assistant text; zero scientific units are produced, so cross-policy transport remains unresolved rather than negative.

**Contribution and closest-work boundary.** The contribution is an identification-and-transport decomposition for reward-conditioned agent memory. First, a byte-identical write intervention plus a stronger reward-preserving wording reduction establishes and broadens the write-time state channel. Second, forced downstream swaps establish conditional behavioral sensitivity without being mislabeled as native retrieval transport. Third, exact retriever, raw-trajectory, endpoint-resolution, and first-action tests localize the negative boundary: broad exposure can coexist with weak reward-branch uptake before terminal evaluation. We make no claim of a new memory architecture, a new reward-robustness method, universally harmful failure memory, a confirmatory generic memory-presence effect, writer- or policy-invariant downstream effects, live-browser transport, or a general law of reward-corruption variance.

## 2 Related Work

**Evaluator reliability.** JudgeBench and RewardBench establish that LLM judges and reward models can be unreliable before their outputs are used for adaptation (Tan et al., 2025; Lambert et al., 2024); Plan-RewardBench exposes additional brittleness on tool-integrated trajectories (Wang et al., 2026a). These works motivate the possibility of a wrong terminal label, but their object ends at scoring accuracy. Our object begins when that label is consumed by a persistent-state update.

**Feedback-conditioned memory construction.** ReasoningBank explicitly distills different memories from successful and failed trajectories (Ouyang et al., 2026); Reflexion stores evaluative verbal feedback for later trials (Shinn et al., 2023); Agent Workflow Memory extracts reusable procedures from interaction histories (Wang et al., 2024). Mem-$\alpha$ learns what to store and update from downstream reward, and AttriMem adds attribution-derived process feedback to learn a memory-construction policy (Wang et al., 2025; Li et al., 2026). We therefore surrender any novelty claim that feedback can shape memory or that success/failure-conditioned writers are new. Our residual treatment fixes the existing construction policy and the source trajectory, then counterfactually changes only which released reward-conditioned branch writes the durable state.

**Raw-experience baselines.** Agent-memory evaluations increasingly include trajectory-retrieval or strong long-context controls rather than omission alone (Ouyang et al., 2026; Xiong et al., 2026; Wang et al., 2026b). We therefore add the exact compact trajectory projection seen by our writer before reward-label conditioning as a native-support comparator, without claiming an exact replication of an external system.

**Memory valuation, reuse, and self-improvement variance.** Memory Reward Inflation studies how imperfect proxy reward can overvalue erroneous memories and amplify them through reuse (Asadolahi et al., 2026). RoMeRL addresses misleading trajectory-level utility updates and the memory-reward trap (Yang et al., 2026). Ye et al. quantify run variance, curriculum sensitivity, and memory underspecification in self-improving agents (Ye et al., 2026). These are the closest downstream neighbors, but their intervention is valuation, utility update, retrieval/reuse, or system-level variation. We instead isolate the earlier write boundary and then carry the resulting state through a matched future-evidence swap. Accordingly, our plug-in variance curve is not presented as a new variance phenomenon or empirical corruption sweep.

**Defended residual.** The narrow novelty is an identification object rather than a new memory algorithm: a byte-identical experience, a reward-conditioned writer-mode intervention under a fixed construction policy, a stronger same-mode wording reduction, and a fully crossed downstream memory swap under fixed future evidence. This distinguishes the paper from both "reward can be wrong" and "feedback can change memory" while keeping the claim local to the released ReasoningBank/WebArena configuration.

## 3 Causal Object: Reward Error as a Persistent-State Intervention

Let task $x$ produce trajectory $\tau$, and let the terminal evaluator emit $r \in \{0, 1\}$. A reward-conditioned memory writer constructs

$$M(r) = G(\tau, r). \tag{1}$$

In the released ReasoningBank implementation, $r = 1$ selects the success-reflection objective and $r = 0$ selects the failure-reflection objective. The intervention studied here holds $\tau$ fixed and changes only this branch. The immediate causal estimand is therefore the paired persistent-state contrast $M(1)$ versus $M(0)$ for the same experience.

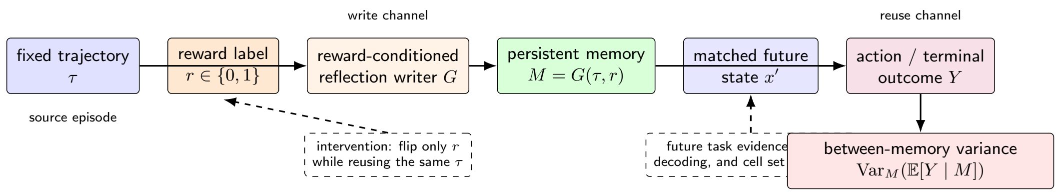


Figure 1: The reward-to-memory-to-outcome channel. The source trajectory is fixed while the reward label selects the reflection writer. The resulting persistent memory is then injected into a matched held-out task packet under a fixed policy configuration. Repeated outcomes identify the between-memory component of run variance.

For a matched future task state $x'$, let $Y$ denote a downstream behavioral or terminal outcome under retrieved memory $M$. The reuse-stage contrast is

$$\Delta_Y(x') = \mathbb{E}[Y \mid do(M = M(0)), x'] - \mathbb{E}[Y \mid do(M = M(1)), x']. \tag{2}$$

The future task evidence and policy configuration are held fixed across the pair. This separates variation introduced by persistent memory state from variation in the observed future state.

### 3.1 A BOUNDED VARIANCE CONSEQUENCE

If terminal-label corruption stochastically selects between the two measured reward-conditioned memories, the resulting state mixture has a direct variance consequence. Let $M$ denote the persistent memory state available to later tasks. The law of total variance gives

$$\mathrm{Var}(Y) = \mathbb{E}_M[\mathrm{Var}(Y \mid M)] + \mathrm{Var}_M(\mathbb{E}[Y \mid M]). \tag{3}$$

The second term is the component induced by variation in persistent memory state. If reward corruption changes which memory is written and the paired memories produce different conditional success probabilities, reward uncertainty contributes directly to this term. Section 7 estimates the paired conditional effect and converts it into an explicit corruption-rate variance curve.

This consequence is secondary to the causal identification above. No parameter update is required for the channel itself: in the released external-memory mechanism, a mislabeled episode can change the context available to a later retrieval.

## 4 EXPERIMENTAL SETUP

**Source trajectories and write estimands.** We use released WebArena Shopping metadata and ReasoningBank trajectories rather than collecting a new self-improvement stream. The original paired write intervention requests six outcome-balanced sources. Four produce complete memory outputs under both reward-conditioned branches; two failure-branch calls terminate at the provider response-state layer before assistant text exists. We neither impute nor re-POST those missing arms, so the original estimate is conditional on paired completion. The prompt-sensitivity control uses eight disjoint trajectories. To test breadth, we deterministically select 16 additional sources before new writer calls, balanced 8/8 by original benchmark outcome and spanning 11 intent-template families; original intervention/control tasks, the four forced-swap futures, and all retrieval-hit tasks reserved by the first exact-retrieval audit are excluded. This breadth sample is fixed independently of new memory or terminal outcomes.

**Models, controls, and gates.** DeepSeek-V4-Flash is the primary memory writer at temperature zero. The original writer surface uses a 2,200-token cap; the breadth parent reveals three failure-branch length-censored responses, so a single preregistered execution repair raises only that cap to 4,096 and regenerates all 32 breadth units fresh. Doubao Seed 2.0 Mini is the downstream terminal policy at temperature 0.2. Write-stage observables are exact content change, token-set Jaccard distance, title-set change, and a fixed operation-slot diagnostic. The strongest write-stage reduction is semantic-preserving same-mode rewording whose lexical perturbation exceeds the released success/failure prompt difference. Terminal tests retain the frozen practical-effect floor 0.15 together with permutation $p < 0.05$; no later experiment lowers this floor after outcomes.

**Forced intervention versus released transport.** The original terminal study deliberately swaps a chosen source memory into four frozen future evidence packets, fully crossing four complete source pairs with four futures. We call this a *forced memory-swap* test: it estimates sensitivity to persistent-state intervention, not retrieval probability. We separately replay the released ReasoningBank retrieval mechanism exactly: all-MiniLM-L6-v2 over task-description documents, current task intent as query, top-1 retrieval, and threshold 0.3. After expanding the source bank to 20 paired sources, the retriever is run over all 812 released WebArena tasks. The transport experiment then freezes every held-out Shopping task that (i) clears the released retrieval threshold, (ii) has a released trajectory from which the fixed browser-state packet can be reconstructed, and (iii) uses a deterministic string-match evaluator. This produces 36 future tasks before terminal outcomes. Each task is automatically bound to its top-1 source; only that source's success- versus failure-conditioned memory is swapped.

**Native-wrapper, omission, and raw-trajectory controls.** For the 36-task transport test, raw memory bytes are content-addressed: the original four source pairs are recovered exactly from historical terminal prompts and checked against their original write SHAs, while the additional 16 pairs come directly from the breadth-replication archive. We serialize the selected memory with the released ReasoningBank human-memory wrapper (retrieved-source preamble plus Title/Description/Content fields), but retain frozen released browser evidence instead of the unavailable live Shopping endpoint. Four fresh rollouts per reward-conditioned branch yield 288 terminal calls. A separately frozen no-memory arm uses the identical 36 tasks, evidence, policy, evaluator, and four-rollout depth, adding 144 calls. The no-memory primary statistic averages the two absolute branch-to-omission differences per task, again requiring practical magnitude at least 0.15 and omnibus permutation $p < 0.05$.

After B5, we freeze a stronger raw writer-input trajectory comparator: the selected source's deterministic compact action/result projection before reward-label conditioning, with no rewritten memory. The same 36 tasks and four-rollout depth add 144 calls; $\frac{1}{2}(|p_S - p_R| + |p_F - p_R|)$ retains the 0.15 floor and permutation $p < 0.05$. This is an internal raw-experience comparator, not an exact external-method replication. A later zero-call fractional-reference re-score is explicitly post-hoc and cannot replace the binary gates. To localize transport before terminal scoring, B10 then freezes released trajectory step 1 for all 36 tasks, strips any pre-existing task-history memory from the state message, and compares success-memory, failure-memory, and no-memory first structured actions with four rollouts each (432 calls). It reuses the legacy F1D action signature—action family plus click index—and its preregistered mean-TV floor 0.20 with a 100,000-draw within-state permutation test; no-memory is secondary branch-location context only.

**Reliability and scope.** All scientific stages are inference-only, with no training or local GPU fine-tuning. Provider outputs are archived before analysis where supported, execution-invalid attempts have zero scientific authority, and missing scientific units are never replaced by favorable tasks. The DeepSeek-V4-Flash downstream-policy transfer is preregistered on the same 36-task support, but its 900-token parent and sole 2,200-token output-cap repair each terminate on the first frozen unit with provider length censoring and no assistant text. This produces zero scientific cross-policy units and is recorded as support failure, not a null effect. Appendix E reports full request accounting and separates scientific non-passes from provider/runtime failures.

## 5 WRITE-TIME INTERVENTION: DOES THE REWARD LABEL CHANGE PERSISTENT MEMORY?

We begin with a paired intervention over released WebArena Shopping trajectories from the Salesforce fragility study. For each source, the task text and released action history are copied byte-for-byte into two ReasoningBank writer calls. One call uses the released success-reflection instruction and the paired call uses the released failure-reflection instruction. DeepSeek-V4-Flash writes memory at temperature zero; task, trajectory, model, sampling, and output format stay fixed.

**Original paired intervention.** Four of six requested trajectory pairs return complete memory outputs in both conditions. The two incomplete requests terminate in the failure-label writer before assistant text is produced and are excluded from the paired calculation. Among the four complete pairs, every pair changes normalized memory content and extracted memory-item titles. Token-set Jaccard distance is 0.691, 0.708, 0.770, and 0.770 for tasks 21, 22, 23, and 25, respectively (mean

0.735). The 4/4 result is therefore an existence result conditional on paired completion. The content changes are procedural as well as lexical: a fixed operation-slot taxonomy changes in all four pairs (mean slot-set distance 0.493). For example, task 21's success-conditioned memory emphasizes entering the review section and preserving direct quotations, whereas its failure-conditioned memory emphasizes exhaustive pagination and verification. This structural diagnostic is descriptive; Section 6 provides the stronger reward-preserving wording reduction.

**Breadth replication over 16 fresh sources.** The small original source count is a natural external-validity concern, so we preregister a disjoint breadth sample before new writer outcomes. The deterministic selection balances eight originally successful and eight originally failed trajectories and first maximizes distinct intent-template families before filling by task ID; it covers 11 template families. Tasks used by the original intervention, prompt control, forced terminal test, and the first retrieval audit's held-out hits are excluded. The parent 2,200-token execution completes 29/32 writer calls but length-censors three failure-branch outputs, so it has no breadth-gate authority. A single execution-only repair raises the output cap to 4,096 uniformly and regenerates all 32 units fresh. All calls then complete with zero provider failures.

All 16 fresh pairs change exact normalized content and title sets. Their mean token-set Jaccard distance is 0.658 (range 0.545–0.850). Combined with the original four complete pairs, the write-time evidence now contains 20/20 divergent paired sources, with pooled mean token-set distance 0.673. Because the breadth sample is outcome-balanced and fixed before these writer outputs, this result strengthens the upstream conclusion without selecting sources for large downstream effects. Figure 2 retains the original pair/control visualization; Appendix D reports the expanded source set and execution receipt.

The write-time conclusion is therefore the strongest part of the evidence chain: under a fixed construction policy, changing which released reward-conditioned branch processes the same experience reliably changes the durable textual state. The next section asks whether this exceeds generic prompt sensitivity; Section 7 then separates forced downstream sensitivity from realized transport through the released retriever.

## 6 Ruling Out Generic Prompt Sensitivity

The reward intervention changes the reflection instruction together with the reward semantics. A generic prompt-sensitivity account therefore predicts that a sufficiently strong wording perturbation inside one reflection mode should produce comparable memory divergence. We test that reduction on eight released trajectories disjoint from the paired write sample and balanced across original outcomes.

Before provider calls, we freeze semantic-preserving rewordings for both reflection modes. Under token-set Jaccard distance, the released success-versus-failure prompts differ by 0.114. The within-success rewording differs by 0.171 and the within-failure rewording by 0.152. The control thus perturbs surface wording more strongly while preserving the reflection objective.

Each fresh trajectory is written with the released success mode and the released failure mode. The same writer also processes one frozen semantic-preserving rewording of each mode at temperature zero. Let $B_i$ be the memory distance between the two released reward modes and let $W_i$ be the mean within-mode rewording distance. The frozen paired statistic is

$$\Delta_{\text{prompt}} = \frac{1}{8}\sum_{i=1}^{8}(B_i - W_i). \tag{4}$$

All 32 control outputs complete. Mean reward-mode distance is 0.700, while the stronger same-mode wording control has mean 0.595. The paired excess is 0.105, and an exact one-sided sign-flip test gives $p = 0.0078$. This passes the frozen 0.10 effect margin together with the $p < 0.05$ gate. The reward-conditioned write effect therefore survives a same-information prompt-wording reduction that is deliberately stronger in lexical perturbation.

The same conclusion is visible at the procedural-structure level. Without changing the operation-slot taxonomy used in the paired write intervention, we apply it to the frozen title sets from all eight

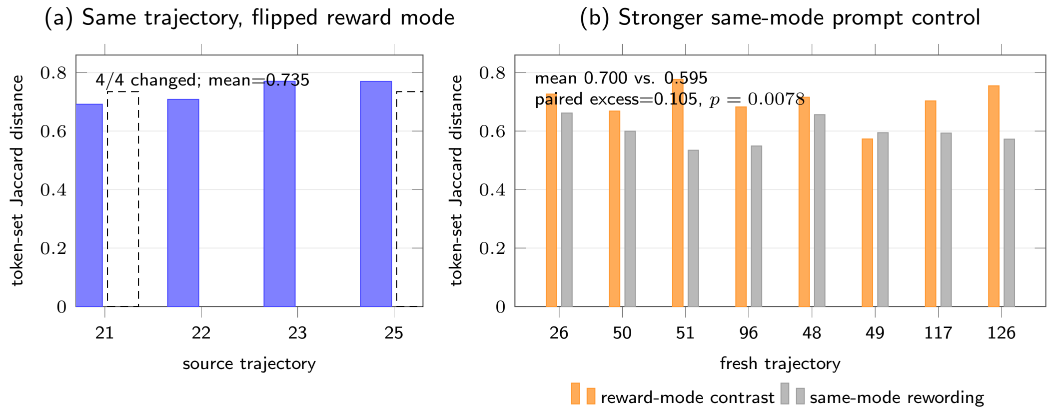


Figure 2: Write-channel evidence and its strongest simple control. (a) Among the four paired write interventions complete in both writer conditions, flipping only the reward-conditioned writer mode changes every memory. (b) On eight fresh control trajectories, reward-mode memory divergence exceeds stronger semantic-preserving same-mode prompt rewording. The control mean is 0.595 versus 0.700 between reward modes; paired excess is 0.105 with exact $p = 0.0078$.

control trajectories. Mean slot-set distance between the released success and failure modes is 0.556, compared with 0.247 for semantic-preserving rewordings within the same mode, an average structural excess of 0.309. Five tasks have positive excess, two tie, and one is negative. We treat this as a descriptive controlled diagnostic, not a second significance test or an embedding-semantic claim. Its role is narrower: the structural shift associated with reward mode is larger on average than the structural shift produced by reward-preserving wording changes under the same fixed taxonomy.

## 7 From Persistent-State Intervention to Realized Transport

A written state difference matters only if it reaches later decisions. We separate two questions that are easy to conflate: *can* paired memories change behavior when experimentally injected, and *does* the released retrieval/interface stack expose those memories broadly enough to produce a practically large terminal effect?

### 7.1 Forced memory swaps establish conditional sensitivity

The original action-reach experiment holds a future browser observation fixed while swapping only the paired persistent memory. Seven of 12 aligned rollouts choose different structured next actions, establishing an action-level existence witness. A stronger distributional stress test with eight rollouts per condition has mean empirical first-action TV distance 0.344 but permutation $p = 0.311$; we therefore do not claim systematic first-action distribution separation.

Terminal sensitivity is tested on all four original complete source-memory pairs and four future tasks frozen before terminal-policy calls. Every source pair is crossed with every future, producing 16 cells under byte-identical future evidence. The initial run uses three rollouts per condition and does not pass its dual gate (mean $|\Delta| = 0.145833$, $p = 0.160128$). Before any confirmatory call, the follow-up contract keeps the same four sources, four futures, 0.15 practical-effect floor, and $p < 0.05$ criterion, forbids source/task selection after outcomes, and changes only uniform replication depth from three to eight. All 256 confirmatory calls complete, yielding mean $|\Delta| = 0.15625$ with permutation $p = 0.00074$. Eight cells are zero; six signed effects are positive and two negative, and the two largest cells contain 78.2% of squared-effect mass. We therefore interpret this result as *forced-intervention sensitivity*: paired persistent states can alter terminal outcomes under matched evidence, but the design does not itself show that the released retriever would select those memories for those futures.

A fresh source-independent no-memory control on the same four futures further rules out a uniform branch ordering. Omission matches both reward-conditioned branches in eight of the 16 source–future comparisons, is closer to the success-conditioned branch in six, and closer to the failure-conditioned branch in two. Thus neither "success memory is always helpful" nor "failure memory is always harmful" explains the forced-swap result.

### 7.2 Exact released-retriever exposure changes the question

ReasoningBank retrieves over task descriptions with all-MiniLM-L6-v2, current intent as query, top-1 selection, and threshold 0.3; reward-conditioned memory content is not part of the retrieval embedding. We replay this released mechanism exactly with the cached model weights and no provider calls. With only the original four source descriptions, 8 of 188 held-out Shopping tasks clear the threshold (4.26%), and none of the four forced-terminal futures does. This establishes a clean distinction: the $4 \times 4$ experiment measures sensitivity to a state intervention, not source-faithful retrieval transport.

The 16-source breadth replication lets us test exposure on a less sparse bank without changing retrieval mechanics. After adding the 16 new source descriptions, exact top-1/0.3 retrieval hits 125 of 172 held-out Shopping tasks (72.7%), spanning 39 intent-template families. Before any new terminal outcomes, we freeze all 36 hits that also have a released trajectory and a deterministic offline string-match evaluator; these cover 13 intent-template families and 11 actually selected source memories. Each future task is bound to the source chosen by the retriever itself.

### 7.3 Broad retrieval-matched transport is a negative boundary

For each of the 36 frozen futures, we recover the selected source's exact success- and failure-conditioned memory bytes and serialize them with the released ReasoningBank human-memory wrapper: the retrieved-source preamble followed by Title/Description/Content fields. Future browser-state evidence, downstream Doubao Seed 2.0 Mini policy, evaluator, temperature, and four-rollout depth remain matched across branches. All 288 calls complete with no provider failures.

The result is sharply different from the forced-swap matrix. Mean absolute success-rate difference is 0.02083 with permutation $p = 0.4289$, far below the unchanged 0.15 practical-effect floor. Thirty-four of 36 futures have zero branch contrast; eighteen are joint 0/0 floors and sixteen joint 1/1 ceilings. Only task 144 (source 141) separates at 0.75 versus 1.0, and task 228 (source 226) at 0.5 versus 0.0. Retrieval similarity itself is not a useful magnitude proxy on this support (descriptive Pearson $r = 0.159$). Thus broad released-retriever exposure does not imply broad terminal sensitivity.

We then omit memory entirely on the identical 36-task support, policy, evidence, evaluator, and four-rollout depth. All 144 calls complete. The preregistered three-arm statistic averages $|p_S - p_0|$ and $|p_F - p_0|$ within task; its mean is 0.04514. An omnibus permutation test gives $p = 0.00147$, but the effect misses the unchanged 0.15 practical floor, so a practically large native memory-presence effect is not established. At the point estimate, no-memory is equidistant from the two reward-conditioned branches on 34/36 tasks, closer to success memory on one, and closer to failure memory on one.

**Where does native transport attenuate?** A 144-call raw writer-input trajectory arm on the same 36 tasks gives mean rewrite-versus-raw effect 0.04514 ($p = 0.00775$), below the 0.15 floor, with 31/36 S/F/raw/no-memory point estimates identical. A post-hoc fractional-reference re-score reduces joint S/F floors from 18 to 10 but leaves mean S/F separation at 0.01951, arguing against binary discretization as the main explanation. We therefore move earlier in the chain. B10 freezes each native-hit task at released trajectory step 1 and samples success-memory, failure-memory, and no-memory first structured actions. All 432 calls complete. Mean S/F first-action TV is 0.06944 with permutation $p = 0.5801$, far below the inherited 0.20 process floor; only 9/36 states have any S/F TV and 0/36 change their modal action. Removing click indices lowers mean S/F action-family TV to 0.02778. A descriptive memory-versus-no-memory first-action TV of 0.17014 with six modal shifts suggests generic procedural context can affect decision-making, but it is not a confirmatory reward-branch effect. Together with the raw and endpoint controls, this localizes the dominant attenuation before or at branch-specific policy uptake rather than solely at terminal scoring.

Table 1: Core identification/transport ladder. Practical-effect floors remain binding when preregistered. Retrieval audits use zero provider calls.

| Stage | Support / calls | Key result | Verdict |
|---|---|---|---|
| Write breadth | 20 complete pairs | 20/20 content/title divergence; pooled Jaccard 0.673 | support |
| Forced terminal swap | 16 cells / 256 | mean $\lvert\Delta\rvert = .15625$, $p = .00074$ | pass |
| Original-bank retrieval | 188 held-out / 0 | 8 hits (4.26%); forced futures 0/4 hit | boundary |
| 20-source retrieval | 172 held-out / 0 | 125 hits (72.7%); 36 terminal-eligible | support |
| Native branch transport | 36 tasks / 288 | mean $\lvert\Delta\rvert = .02083$, $p = .4289$ | non-pass |
| Native controls | 36 tasks / 288 | omission .04514 ($p = .00147$); raw .04514 ($p = .00775$) | both floor fail |
| Native first-action uptake | 36 states / 432 | S/F TV .06944, $p = .5801$; 0/36 modal diff. | non-pass |

### 7.4 What generalizes, and what does not?

A GLM-5.3 writer replication changes all four original paired memories (mean token-set Jaccard 0.737), but its matched 256-call terminal effect is 0.140625 ($p = 0.00012$), below the frozen 0.15 floor, with two sign reversals. A preregistered DeepSeek-V4-Flash policy transfer remains unresolved: its first frozen unit length-censors without assistant text at both the 900-token parent and sole allowed 2,200-token repair, so the line stops with zero scientific units.

Overall, forced swaps establish conditional leverage, native retrieval establishes exposure, raw/endpoint controls reject two simple explanations for the broad terminal non-pass, and B10 places the dominant attenuation before or at reward-branch-specific policy uptake. Appendix D gives cell-level results and bounded consequence analyses.

## 8 Limitations

The original six-source intervention has two failure-branch provider incompletions, so its 4/4 estimate is conditional on paired completion. The disjoint 16/16 breadth replication reduces the source-count concern but remains one released Shopping collection. GLM-5.3 reproduces upstream divergence on four sources but misses the terminal practical-effect floor; writer-invariant effects are not established.

The forced $4 \times 4$ futures are not original-bank retrieval hits, and native transport still uses frozen browser-state packets rather than live endpoints. On the 36-task support, terminal S/F separation is 0.02083; no-memory/raw controls are 0.04514; fractional-reference separation remains 0.01951. B10 likewise fails at first action (TV 0.06944, $p = 0.5801$, 0/36 modal S/F differences). The larger descriptive memory-versus-no-memory TV (0.17014) is not confirmatory. We therefore localize attenuation to branch-specific uptake on this fixed-state interface without claiming live or population transport.

DeepSeek cross-policy transfer remains a provider-support stop, not a scientific null. Claims are local to the released ReasoningBank top-1/0.3 substrate; learned retrieval, multi-memory composition, and reward-dependent membership are different substrates. Token Jaccard, operation slots, and corruption-rate calculations remain bounded diagnostics.

## 9 Conclusion

Terminal feedback in reward-conditioned external memory is also a state-update input. Under byte-identical experiences, 20/20 complete paired sources produce different persistent memories and a stronger reward-preserving wording control does not absorb the effect. Forced swaps show conditional terminal sensitivity (mean $|\Delta| = 0.15625$, $p = 0.00074$), while exact retrieval audits prevent equating intervention sensitivity with realized exposure.

After expansion, 125/172 held-out Shopping tasks clear the released threshold, yet the 36-task native terminal S/F contrast is only 0.02083. Matched no-memory and raw writer-input controls each yield 0.04514 mean absolute effect, and a post-hoc fractional-reference diagnostic reveals no hidden large branch separation. More decisively, the same 36-task support yields first-action S/F TV 0.06944 ($p = 0.5801$), with identical modal actions in every state. Reward errors robustly change what is written, but reward-branch-specific uptake is weak on broad native-hit futures. Realized impact is therefore gated sequentially by write, exposure, uptake, and outcome transport; downstream risk cannot be inferred from memory divergence alone.

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## A F0 Reproduction Details

### A.1 Released trajectory sample

The F0 sample uses six task IDs from the released AWM shuffle-seed-42 Shopping parquet: 21 through 25, plus 47. The deterministic sampler selects the lowest task IDs with parseable trajectory records within each original-outcome stratum. Original failures are 21 through 22 plus 24. Original successes are 23 plus 25 and 47.

### A.2 Memory construction intervention

Each released trajectory is summarized into an action trace. The same trace is inserted into two prompts. One prompt contains the released ReasoningBank success-reflection system instruction. The paired prompt contains the released failure-reflection system instruction. The task prompt stays fixed. The writer model stays fixed. Temperature stays at zero. The raw provider output is archived before metric computation.

### A.3 Pair-level F0 results

Table 2: Complete paired interventions.

| Task | Original outcome | Content changed | Token distance |
|---|---|---|---|
| 21 | failure | yes | 0.691489 |
| 22 | failure | yes | 0.708108 |
| 23 | success | yes | 0.769953 |
| 25 | success | yes | 0.769608 |

The extracted memory-item title set changes for every complete pair. The mean token-set Jaccard distance is 0.734789. A deterministic post-ready strategy-slot audit maps the frozen titles into navigation, pagination/coverage, query expansion, semantic matching, evidence capture, extraction, and verification/completeness. All four complete pairs change their slot set. The slot-set Jaccard distances are 0.571, 0.400, 0.500, and 0.500 for tasks 21, 22, 23, and 25, respectively, with mean 0.493. This is a structural diagnostic over frozen titles, not an embedding-based semantic claim.

Provider failures for task 24 and task 47 are excluded from the paired analysis. Both are failure-label calls with `ArkResponseStateError`; neither produces assistant text or a raw memory SHA. Private response identifiers are retained and GET-only recovery finds no text. The four completed failure-label calls produce 777, 852, 1,113, and 1,296 output tokens. These receipts distinguish provider response-state failures from a later memory-parser rejection, while leaving arm-dependent completion bias unresolved.

## B Prompt-Perturbation Control Details

The control set is disjoint from the F0 requests. Each fresh trajectory is written under both released reward modes. One semantic-preserving rewording is also evaluated within each mode. Table 3 reports the paired distances used in the frozen sign-flip test.

Table 3: Full fresh prompt-perturbation control. Between is the released success-versus-failure memory distance; Within averages the two same-mode rewording distances.

| Task | Original outcome | Between | Within | $B_i - W_i$ |
|---|---|---|---|---|
| 26 | failure | 0.726 | 0.662 | 0.065 |
| 50 | failure | 0.668 | 0.599 | 0.069 |
| 51 | failure | 0.776 | 0.534 | 0.242 |
| 96 | failure | 0.682 | 0.549 | 0.134 |
| 48 | success | 0.716 | 0.656 | 0.059 |
| 49 | success | 0.573 | 0.594 | -0.022 |
| 117 | success | 0.703 | 0.593 | 0.110 |
| 126 | success | 0.755 | 0.572 | 0.182 |
| Mean | | 0.700 | 0.595 | 0.105 |

## C Downstream Confirmatory Details

### C.1 Fixed-state action distributions

The distributional action stress test retains all four intervention states from the action-reach experiment and uses eight rollouts per memory condition. The per-state success-memory versus failure-memory TV distances are 0.625 for task 22 and 0.375 for task 23. They are 0.125 for task 24 and 0.250 for task 26. Their mean is 0.34375. The frozen 100,000-draw permutation audit gives $p = 0.310527$.

### C.2 Frozen terminal matrix

The initial terminal experiment and F2R1 use the same complete F0 source set $\{21, 22, 23, 25\}$, the same future-task set $\{164, 385, 387, 388\}$, and the same aggregate absolute-effect statistic with the same dual gate: mean absolute difference at least 0.15 and within-cell permutation $p < 0.05$.

The initial stage uses three fresh rollouts per condition per cell and is already frozen in Git commit `730c2bc` at 2026-08-22 00:01:42 +08:00; it yields 0.145833 with $p = 0.160128$ and does not pass. The separately authorized F2R1 contract is content-addressed as `ea9a2260...01f5`; its first provider receipt is generated at 2026-08-22 01:50:32 +08:00. It explicitly forbids source/future-task selection after outcomes and increases only uniform replication depth to eight fresh rollouts per condition in every cell. All 256 F2R1 calls complete, yielding 0.15625 with $p = 0.00074$. Thus F2R1 is a targeted same-support replication after an initial non-pass, not a claim that the confirmatory design was chosen before any terminal outcomes were observed.

Table 4: Initial F2 and F2R1 cell-level terminal rates on identical 4-by-4 support. S/F denote success-/failure-label memory.

| Source | Future | F2 S | F2 F | F2 $\lvert\Delta\rvert$ | F2R1 S | F2R1 F | F2R1 $\lvert\Delta\rvert$ |
|---|---|---|---|---|---|---|---|
| 21 | 164 | 1.000 | 1.000 | 0.000 | 1.000 | 1.000 | 0.000 |
| 21 | 385 | 0.000 | 0.333 | 0.333 | 0.000 | 0.250 | 0.250 |
| 21 | 387 | 0.000 | 0.333 | 0.333 | 0.125 | 0.250 | 0.125 |
| 21 | 388 | 1.000 | 0.333 | 0.667 | 1.000 | 0.625 | 0.375 |
| 22 | 164 | 1.000 | 1.000 | 0.000 | 1.000 | 1.000 | 0.000 |
| 22 | 385 | 0.000 | 0.667 | 0.667 | 0.000 | 0.125 | 0.125 |
| 22 | 387 | 0.000 | 0.000 | 0.000 | 0.125 | 0.000 | 0.125 |
| 22 | 388 | 1.000 | 1.000 | 0.000 | 0.250 | 1.000 | 0.750 |
| 23 | 164 | 1.000 | 1.000 | 0.000 | 1.000 | 1.000 | 0.000 |
| 23 | 385 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| 23 | 387 | 0.000 | 0.000 | 0.000 | 0.125 | 0.250 | 0.125 |
| 23 | 388 | 0.667 | 1.000 | 0.333 | 1.000 | 1.000 | 0.000 |
| 25 | 164 | 1.000 | 1.000 | 0.000 | 1.000 | 1.000 | 0.000 |
| 25 | 385 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| 25 | 387 | 0.000 | 0.000 | 0.000 | 0.125 | 0.750 | 0.625 |
| 25 | 388 | 1.000 | 1.000 | 0.000 | 1.000 | 1.000 | 0.000 |
| Mean $\lvert\Delta\rvert$ | | 0.145833 | | | 0.15625 | | |

## C.3 Fresh no-memory branch-location control

The no-memory control is frozen after F2R1 as a secondary branch-location diagnostic. It uses the identical four future tasks, released evidence packets, Doubao Seed 2.0 Mini settings, evaluator, and eight-rollout depth, but no source-memory dimension. The earlier 12 exploratory no-memory calls are excluded. An initial 32-call execution is also excluded from science because a new local validator incorrectly required the provider's resolved model name to equal the requested alias; all 32 responses resolved to `doubao-seed-2-0-mini-260215`, which is exactly the versioned model name recorded in all 256 historical F2R1 receipts. After freezing that execution-layer diagnosis and archiving provider responses before local validation, the one-for-one recovery completes 32/32 calls. Both attempts are counted in execution cost; only the recovery outcomes enter the analysis.

Table 5: Fresh no-memory terminal baseline. Wilson intervals are per future task and are not replicated across source memories.

| Future task | Successes | Rate | Wilson 95% interval |
|---|---|---|---|
| 164 | 8/8 | 1.000 | [0.676, 1.000] |
| 385 | 0/8 | 0.000 | [0.000, 0.324] |
| 387 | 0/8 | 0.000 | [0.000, 0.324] |
| 388 | 8/8 | 1.000 | [0.676, 1.000] |

Combining these four source-independent baselines descriptively with the 16 F2R1 cells yields eight cells where omission matches both memory branches, six where omission is closer to the success-memory branch, and two where it is closer to the failure-memory branch. The largest illustrative contrasts are source/future 22/388, where $(p_S, p_F, p_0) = (0.25, 1.0, 1.0)$, and 25/387, where $(p_S, p_F, p_0) = (0.125, 0.75, 0.0)$. The first isolates the success-conditioned memory as the deviation from omission; the second places both memories above omission. Because each $p_0$ is shared by four source rows, we report cellwise score intervals and no new global significance test.

### C.4 CROSS-WRITER REPLICATION AND FROZEN NEGATIVE BOUNDARY

The writer-family test is sequential. Stage 1 first rewrites only source tasks 21, 22, 23, and 25 under the same released success/failure ReasoningBank prompts using GLM-5.3 at temperature zero. The parent 2,200-token attempt is scientifically invalid after two failure-mode responses terminate at the exact output cap with no assistant text; an overlapping-runner race also makes its provider POST count reconstructible only as a lower bound of nine. The single preregistered repair changes only the output cap to 4,096 and adds a single-writer transaction lock. Its eight calls all complete; all four pairs change content and title sets, with token-set distances 0.783, 0.728, 0.715, and 0.723 (mean 0.737482).

Stage 2 keeps the original Doubao Seed 2.0 Mini downstream policy, four future tasks, evidence packets, evaluator, eight-rollout cell depth, and dual terminal gate unchanged. All 256 cells-by-condition rollouts are scorable. The observed mean absolute effect is 0.140625 and the 100,000-draw permutation test gives $p = 0.00012$. The statistical gate passes but the frozen 0.15 minimum-effect gate does not, so the registered cross-writer terminal generalization claim is not established. Table 6 shows that the difference is not a uniform attenuation.

Table 6: Signed failure-minus-success terminal effects under the original DeepSeek-V4-Flash writer (D) and the GLM-5.3 writer (G).

| Source | Future | D effect | G effect |
|---|---|---|---|
| 21 | 164 | 0.000 | 0.000 |
| 21 | 385 | 0.250 | 0.250 |
| 21 | 387 | 0.125 | 0.375 |
| 21 | 388 | -0.375 | 0.000 |
| 22 | 164 | 0.000 | 0.000 |
| 22 | 385 | 0.125 | 0.000 |
| 22 | 387 | -0.125 | -0.125 |
| 22 | 388 | 0.750 | -0.250 |
| 23 | 164 | 0.000 | 0.000 |
| 23 | 385 | 0.000 | 0.000 |
| 23 | 387 | 0.125 | -0.750 |
| 23 | 388 | 0.000 | 0.000 |
| 25 | 164 | 0.000 | 0.000 |
| 25 | 385 | 0.000 | 0.000 |
| 25 | 387 | 0.625 | 0.500 |
| 25 | 388 | 0.000 | 0.000 |

Among the six cells that are nonzero under both writers, four retain direction and two reverse. Reusing the already frozen writer-independent no-memory rates only as descriptive baselines yields 10 GLM cells equidistant from omission, four closer to the success-memory branch, and two closer to the failure-memory branch; no additional provider calls or global test are introduced. This supports the upstream write-channel replication while preserving the failed terminal-invariance gate.

### C.5 HETEROGENEITY AND CORRUPTION-RATE DECOMPOSITION

For each frozen cell, the confirmatory run estimates paired conditional success probabilities. Eight of the 16 cells have zero observed success-rate contrast. Six have positive failure-memory minus success-memory effects and two have negative effects. The largest squared effects occur at source/future pairs 22/388 and 25/387; together they contribute 78.2051% of the total squared-effect mass. This sign and magnitude heterogeneity is why the paper treats the result as a variance channel instead of a uniform directional harm effect.

If reward-label corruption selects the failure-conditioned memory with probability $q$, the induced between-memory variance equals $q(1-q)\Delta_c^2$ in cell $c$. The exact mean squared conditional effect across all 16 cells is 0.076172. Averaging over cells gives the curve $0.076172q(1-q)$, with value 0.006855 at $q = 0.1$, value 0.014282 at $q = 0.25$, and value 0.019043 at $q = 0.5$.

For finite-cell sensitivity, we run a fixed-seed nonparametric bootstrap with 100,000 repetitions, resampling 16 source–future cells with replacement. The percentile 95% interval for mean absolute effect is $[0.054688, 0.273438]$. The corresponding interval for mean squared effect is $[0.010742, 0.164062]$, which maps to $[0.002686, 0.041016]$ for $V(0.5)$. This bootstrap is descriptive sensitivity to the observed cell composition. The preregistered within-cell permutation test on the 256 confirmatory rollouts supplies the primary significance result.

### C.6 Controlled structural diagnostic

The controlled structural diagnostic reuses the exact operation-slot taxonomy already frozen for the F0 title audit; it does not add a learned semantic judge. Applied to the eight disjoint prompt-control trajectories, the mean slot-set distance between released reward modes is 0.555655, whereas the mean distance induced by semantic-preserving rewording within the same mode is 0.246578. The mean between-minus-within structural excess is 0.309077; five tasks are positive, two tie, and one is negative. This is descriptive supporting evidence rather than a second hypothesis test. On the four complete F0 pairs, success-only operation slots include at least one of evidence capture, navigation, or query expansion in all four pairs, while verification/completeness is failure-only in three pairs. Provider receipts also show longer failure-mode responses in all four complete pairs (mean output-token difference +266.5; mean reasoning-token difference +268.0), which we report only as response-structure asymmetry, not as semantic quality.

### C.7 Source–future interaction diagnostic

A two-way additive decomposition of the centered signed F2R1 cell effects is shown below. It is a finite-matrix description, not inferential ANOVA.

Table 7: Descriptive decomposition of centered signed terminal effects.

| Component | Sum of squares | Share |
|---|---|---|
| Source-memory main effect | 0.101562 | 9.4% |
| Future-task main effect | 0.070312 | 6.5% |
| Source $\times$ future interaction residual | 0.906250 | 84.1% |
| Total | 1.078125 | 100% |

The source and future marginals therefore summarize little of the signed effect structure. Source 21 is positive on future tasks 385 and 387 but negative on 388; source 22 is positive on 385 and 388 but negative on 387. Future tasks 387 and 388 likewise change sign across source memories. Table 8 adds benchmark-observable future-task attributes. Future task 164 is at a joint 1.0/1.0 terminal ceiling

Table 8: Future-task attributes and concentration of the frozen terminal effect. "Home" means the task starts at the Shopping root rather than a product page.

| Task | Start | Ref. items | Mean $\|\Delta\|$ | Squared-mass share |
|---|---|---|---|---|
| 164 | Product | 2 | 0.00000 | 0.0% |
| 385 | Home | 8 | 0.09375 | 6.4% |
| 387 | Home | 6 | 0.25000 | 35.9% |
| 388 | Home | 2 | 0.28125 | 57.7% |

for every source pair, so it has no observable binary-outcome headroom and contributes zero effect mass. The other three tasks contain all observed squared-effect mass. With only four future tasks, start context, task identity, and outcome headroom are confounded; these rows characterize where the frozen effect occurred but do not define a general transfer predictor.

A second descriptive check asks whether sources whose paired memories differ more at write time simply create larger downstream effects. Table 9 shows that this reduction is not supported by the four completed sources. Sources 23 and 25 are effectively matched on both reported write-divergence magnitudes but differ fivefold in source-averaged terminal sensitivity. Across the four points, the descriptive Pearson correlations are $-0.730$ for token distance and $-0.367$ for slot distance versus mean terminal $|\Delta|$. With $n = 4$ these correlations have no inferential status; they are included only to reject the simple monotonic reading that a larger textual or operation-slot perturbation necessarily produces a larger downstream effect.

Table 9: Write-stage divergence versus source-averaged terminal sensitivity. The final column averages $|\Delta|$ over the four frozen future tasks.

| Source | Token distance | Slot distance | Mean terminal $\lvert\Delta\rvert$ |
|---|---|---|---|
| 21 | 0.691 | 0.571 | 0.188 |
| 22 | 0.708 | 0.400 | 0.250 |
| 23 | 0.770 | 0.500 | 0.031 |
| 25 | 0.770 | 0.500 | 0.156 |

# D Breadth, Retrieval, and Transport Expansion

## D.1 Sixteen-source write replication

The breadth sample is selected before new writer outputs. It excludes all original write/control tasks, the four forced-terminal futures, and the eight held-out tasks identified as retrieval hits by the first exact-retriever audit. Within each original-outcome stratum, selection first takes the lowest task ID from each previously unused intent-template family, then fills remaining slots by task ID. Table 10 reports all 16 fresh pairs. The 2,200-token parent run produces three failure-branch length-censored calls (tasks 118, 125, and 232); its 29 completed units are not used for the breadth gate. The sole repair changes only the output cap to 4,096, regenerates all 32 units fresh, and completes every pair.

Table 10: Fresh source-breadth replication. O is the trajectory's original benchmark outcome; F/S denote failure/success. All 16 pairs change both exact content and memory-item title sets.

| Task | O | Jaccard | Task | O | Jaccard |
|---|---|---|---|---|---|
| 118 | F | .850 | 145 | S | .594 |
| 125 | F | .693 | 148 | S | .594 |
| 141 | F | .678 | 159 | S | .732 |
| 146 | F | .565 | 188 | S | .682 |
| 158 | F | .668 | 225 | S | .587 |
| 163 | F | .642 | 226 | S | .732 |
| 189 | F | .699 | 231 | S | .545 |
| 232 | F | .550 | 260 | S | .712 |

The fresh mean token-set distance is 0.65757. Pooling these 16 complete pairs with the four original complete pairs gives 20/20 exact-content changes, 20/20 title-set changes, and pooled mean token-set distance 0.673014.

## D.2 Exact released-retriever exposure

The retrieval audit directly executes the cached all-MiniLM-L6-v2 encoder with the released SentenceTransformer mean-pooling and normalization configuration, maximum sequence length 256, top-1 retrieval, and threshold 0.3. No provider call is used. Table 11 separates exposure under the original sparse bank from exposure after source expansion.

Table 11: Exact released-retriever exposure on held-out Shopping tasks. "Eligible" additionally requires a released trajectory and deterministic offline string-match evaluator.

| Memory bank | Held-out | Hits | Hit rate | Eligible |
|---|---|---|---|---|
| 4 original sources | 188 | 8 | 4.26% | – |
| 20 sources | 172 | 125 | 72.67% | 36 |

None of the original four forced-terminal futures retrieves an original-bank memory. After expansion, future task 164 retrieves new source 163 with cosine 1.0; the other three original futures remain below threshold. Across the 20-source bank, the 125 hits span 39 intent-template families; the 36 pre-outcome terminal-eligible tasks span 13 families and retrieve 11 distinct sources.

### D.3 FULL 36-TASK NATIVE-WRAPPER OUTCOMES

Table 12 exposes all frozen transport cells. S/F are success- and failure-conditioned retrieved-memory rates; N is the independently generated no-memory rate. Each rate uses four fresh Doubao Seed 2.0 Mini rollouts. The reward-branch experiment has mean $|S - F| = 0.020833$ and permutation $p = 0.428866$. The three-arm memory-presence statistic has mean 0.045139 and omnibus $p = 0.00147$, but misses the frozen 0.15 practical floor. Only tasks 144, 228, and 319 have nonzero branch-to-omission magnitude; 34/36 tasks have S=F, comprising 18 joint 0/0 floors and 16 joint 1/1 ceilings.

Table 12: All 36 pre-outcome retrieval-matched fixed-evidence tasks. Src is the exact released top-1 source. P is the mean absolute memory-presence effect $\frac{1}{2}(|S - N| + |F - N|)$.

| Task | Src | S | F | N | P | Task | Src | S | F | N | P |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 26 | 25 | 0 | 0 | 0 | 0 | 229 | 226 | 1 | 1 | 1 | 0 |
| 124 | 226 | 0 | 0 | 0 | 0 | 230 | 226 | 1 | 1 | 1 | 0 |
| 126 | 226 | 1 | 1 | 1 | 0 | 233 | 232 | 1 | 1 | 1 | 0 |
| 142 | 141 | 0 | 0 | 0 | 0 | 279 | 226 | 0 | 0 | 0 | 0 |
| 143 | 141 | 0 | 0 | 0 | 0 | 280 | 226 | 0 | 0 | 0 | 0 |
| 144 | 141 | .75 | 1 | 0 | .875 | 281 | 225 | 0 | 0 | 0 | 0 |
| 147 | 146 | 0 | 0 | 0 | 0 | 282 | 226 | 0 | 0 | 0 | 0 |
| 149 | 148 | 1 | 1 | 1 | 0 | 319 | 188 | 1 | 1 | .5 | .5 |
| 150 | 226 | 1 | 1 | 1 | 0 | 320 | 188 | 0 | 0 | 0 | 0 |
| 164 | 163 | 1 | 1 | 1 | 0 | 321 | 188 | 0 | 0 | 0 | 0 |
| 165 | 163 | 0 | 0 | 0 | 0 | 322 | 188 | 1 | 1 | 1 | 0 |
| 167 | 163 | 0 | 0 | 0 | 0 | 323 | 188 | 0 | 0 | 0 | 0 |
| 190 | 189 | 1 | 1 | 1 | 0 | 329 | 141 | 1 | 1 | 1 | 0 |
| 192 | 188 | 0 | 0 | 0 | 0 | 330 | 141 | 0 | 0 | 0 | 0 |
| 227 | 226 | 1 | 1 | 1 | 0 | 331 | 141 | 1 | 1 | 1 | 0 |
| 228 | 226 | .5 | 0 | 0 | .25 | 333 | 141 | 0 | 0 | 0 | 0 |
| 358 | 189 | 1 | 1 | 1 | 0 | 360 | 231 | 1 | 1 | 1 | 0 |
| 362 | 231 | 1 | 1 | 1 | 0 | 384 | 25 | 0 | 0 | 0 | 0 |

### D.4 RAW WRITER-INPUT TRAJECTORY BASELINE

A stronger native-support comparator replaces the rewritten memory with the compact action/result trace that the C1 writer itself receives before counterfactual reward-label conditioning. Source identity is still selected by the exact released top-1/0.3 retriever. No counterfactual success/failure label and no rewritten memory item are supplied. Future evidence, Doubao Seed 2.0 Mini, temperature, evaluator, and four-rollout depth are identical to B4/B5. All 144 calls complete with zero provider failures. The preregistered statistic $\frac{1}{2}(|S - R| + |F - R|)$ averages 0.045139; a 100,000-repetition three-arm permutation test gives $p = 0.00775$, but the unchanged 0.15 practical floor is missed. This is therefore a statistical deviation without a practically large rewrite-versus-raw effect on the frozen support.

Thirty-one of 36 tasks have identical success-memory, failure-memory, raw-trajectory, and no-memory point estimates, and raw equals no-memory on 32/36. Table 13 lists the only five tasks where raw differs from at least one other arm. A derived tie-aware audit is used for geometry only: 34/36 tasks place raw equally close to success memory, failure memory, and no-memory; task 144 is uniquely closest to failure memory, and task 228 ties failure memory with no-memory. We exclude the runner's single-label "closest arm" secondary field because its deterministic lexical tie-break collapses these ties; this reporting correction does not affect the primary B8 statistic, permutation test, or verdict.

Table 13: The five native-support tasks where the raw writer-input trajectory rate R differs from at least one other point estimate. S/F are reward-conditioned memories and N is omission.

| Task | Src | S | F | R | N |
|---|---|---|---|---|---|
| 144 | 141 | .75 | 1 | 1 | 0 |
| 228 | 226 | .5 | 0 | 0 | 0 |
| 230 | 226 | 1 | 1 | .5 | 1 |
| 319 | 188 | 1 | 1 | .75 | .5 |
| 331 | 141 | 1 | 1 | .5 | 1 |

### D.5 Post-hoc endpoint-headroom diagnostic

Sixteen of the 36 native-support tasks require more than one reference substring. The released binary evaluator multiplies these checks, so missing any required item maps the rollout to zero. After observing B4/B5 saturation, we therefore re-score the already archived 432 success/failure/no-memory outputs with fractional reference coverage. This introduces zero provider calls and is explicitly diagnostic-only: it has no confirmatory gate, does not alter the released evaluator, and cannot retroactively rescue B4/B5.

The diagnostic confirms some endpoint compression but not a hidden large reward-branch effect. Joint success/failure floors fall from 18 under the binary score to 10 under fractional coverage. Nevertheless, mean absolute S/F separation is 0.019511 over all 36 tasks and 0.028274 over the frozen 16-task multi-reference subset. Mean absolute memory-presence separation under fractional coverage is 0.045635 over all 36 tasks. Only three tasks have identical binary S/F means but different fractional-coverage S/F means. We therefore retain endpoint headroom as one transport gate without using score resolution to explain away the native branch non-pass.

### D.6 Native first-action transport localization

To test whether terminal attenuation occurs only after decisions are made, we freeze a pre-terminal process endpoint on the identical 36 native-hit tasks. For every task, the selected source remains the exact released top-1 source. The future observation is the released trajectory's step-1 browser state, with any pre-existing task-history memory stripped before the experiment; there is no task-specific step selection. Success-memory, failure-memory, and no-memory conditions each receive four fresh Doubao Seed 2.0 Mini rollouts at the same temperature and output cap as B4/B5. The action signature is inherited from the earlier F1D protocol: first structured action name, with click-element index retained for clicks. All 432 calls complete with zero provider or unrecoverable parse failures.

The preregistered statistic is mean per-state empirical TV between success- and failure-memory first-action distributions, with the inherited 0.20 process floor and 100,000-draw within-state permutation test. The observed mean TV is 0.069444 with $p = 0.580094$, so branch-specific first-action transport is not established. Nine of 36 states have any nonzero S/F TV, but none changes its modal S/F action. Coarsening clicks by removing element indices reduces mean S/F action-family TV to 0.027778 and leaves only four states with any family-level difference. The secondary no-memory geometry is larger but descriptive: mean $\frac{1}{2}(\mathrm{TV}(S, N) + \mathrm{TV}(F, N)) = 0.170139$, and six states have a modal action under at least one memory branch that differs from no-memory. Because no confirmatory presence gate was preregistered, this supports only the interpretation that generic procedural context can influence action selection while reward-branch-specific uptake remains weak.

A zero-call post-hoc next-goal diagnostic uses the archived B10 outputs without adding provider calls. Mean cross-branch token-set Jaccard distance between success and failure next-goal texts is 0.464370, while the average within-branch rollout distance is already 0.447776; the branch-specific excess is therefore only 0.016593. By contrast, mean memory-versus-no-memory next-goal distance is 0.554403. This diagnostic has no inferential authority and cannot rescue the B10 gate; it further localizes the observed attenuation before or at branch-specific operative planning rather than solely at terminal scoring.

Table 14: B10 process localization on the 36 frozen native-hit states. Presence TV is $\frac{1}{2}(\mathrm{TV}(S, N) + \mathrm{TV}(F, N))$.

| Metric | Value | Registered status |
|---|---|---|
| Mean S/F first-action TV | 0.069444 | floor fail (0.20) |
| Permutation $p$ | 0.580094 | fail |
| States with nonzero S/F TV | 9/36 | descriptive |
| States with modal S/F action difference | 0/36 | descriptive |
| Mean coarse action-family S/F TV | 0.027778 | post-hoc |
| Mean memory-presence first-action TV | 0.170139 | secondary |
| States with memory/no-memory modal shift | 6/36 | secondary |

### D.7 CROSS-POLICY SUPPORT STOP

The downstream-policy transfer is frozen on the identical 36-task support before any DeepSeek-V4-Flash outcome. At the Doubao-matched 900-token cap, the first frozen unit ends with provider status incomplete, reason `length`, and no assistant text. One execution-only repair raises the cap uniformly to 2,200, requires all 288 units to be fresh, and forbids a second repair; the same first unit again length-censors without assistant text. The cross-policy line therefore stops after two provider POSTs with zero scientific units. The preregistered no-memory continuation is not triggered because its condition is complete paired execution, regardless of paired scientific outcome.

### D.8 BOUNDED CORRUPTION-RATE CONSEQUENCE

For completeness, the earlier forced-swap matrix implies a descriptive between-memory variance under a hypothetical reward-label corruption probability $q$. If $p_S(c)$ and $p_F(c)$ are the forced-swap conditional success probabilities in cell $c$, then $V_c(q) = q(1-q)[p_F(c) - p_S(c)]^2$. The mean squared conditional effect across the 16 cells is 0.076172, giving $\bar{V}(q) = 0.076172q(1-q)$ and 0.019043 at $q = 0.5$. A 100,000-repetition cell bootstrap gives a descriptive $V(0.5)$ interval $[0.002686, 0.041016]$. This is a deterministic consequence analysis of the forced intervention, not an empirical corruption sweep or a claim about native transport.

## E EXECUTION ACCOUNTING AND RELIABILITY

All scientific evidence is inference-only; there are no training runs or local GPU fine-tuning runs. Before the baseline-aligned expansion, the observable provider-POST lower bound was at least 841, excluding the unresolved low-level request count of the early 12-unit aligned action witness. The original expansion adds 498 observable POSTs: 32 breadth-parent writer calls, 32 fresh breadth-repair writer calls, 288 native retrieval-matched branch rollouts, 144 matched no-memory rollouts, and two DeepSeek cross-policy support-failure calls. The raw writer-input trajectory follow-up adds 144 terminal POSTs, and the native first-action localization adds 432 process-level POSTs. Across expansion, raw follow-up, and B10 this is 1,074 observable POSTs, of which 1,040 are scientifically usable completions: 32 breadth writer calls, 576 terminal rollouts, and 432 first-action rollouts. The updated full-paper observable POST lower bound is therefore at least 1,915. Retrieval audits, endpoint-headroom/tie-aware analyses, and the B10 next-goal diagnostic use zero provider calls.

The 32-call breadth parent completes 29 units but length-censors three failure-branch writes; only the fresh uniform 4,096-token repair has breadth authority. B4, B5, B8, and B10 complete 288/288, 144/144, 144/144, and 432/432 provider calls, respectively, with zero provider failures in those scientific runs; B10 also has zero unrecoverable parse failures. The DeepSeek policy-transfer parent and sole output-cap repair each fail on their first unit with no assistant text and contribute zero scientific units. Earlier execution failures—the two original F0 failure-arm incompletions, the execution-invalid first no-memory attempt, and the failed parent GLM writer attempt—remain recorded separately and are never imputed as scientific outcomes. Provider outputs are content-addressed or response-first archived where the frozen runner supports it.